

Hall Ticket No.

Code: R122132

Stanley College of Engineering & Technology for Women (A)

B.E III Semester (Main) Examinations - January-2023

ECE & EEE-Probability Theory & Stochastic Processes

Time: 03 Hours

Max. Marks-60

- Note: i). First Question is Compulsory. Answer any Four out of remaining Six questions.
ii) Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii) Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Define the Probability? What are the axioms of probability? [2M CO1 BTL1]
b. Find out the probability of getting a 2 or a 5 when a die is rolled? [2M CO3 BTL1]
c. State the Central Limit Theorem with an example [2M CO2 BTL2]
d. Summarize the different measures of Central Tendency [2M CO3 BTL2]
e. Distinguish between ensemble averages and the time averages of random process? [2M CO4 BTL2]
f. Write the equations of Wiener-Khinchine relations? [2M CO5 BTL2]

Part-B

4X12=48M

2. a) State and prove the Bayes' theorem and Total probability Theorem? [8M CO1 BTL2]
b) When two dice are thrown, determine the probabilities from axiom 3 for the following events. (i) $A = \{\text{Sum} = 7\}$ (ii) $B = \{8 \leq \text{Sum} \leq 11\}$ (iii) $C = \{10 < \text{Sum}\}$ (iv) $P(B \cap C)$ [4M CO1 BTL3]
3. a) Define the CDF? State and prove its properties [6M CO1 BTL4]
b) A continuous random variable X defined by probability density function given by $f(x) = 5(1-x^4)/4$ $0 \leq x \leq 1$. Calculate $E[X]$, $E[X^2]$ and variance. [5M CO2 BTL3]
4. a) Explain how Moment generating function generates the moments? State and prove its properties? [7M CO2 BTL4]
b) The joint probability density function of $f(x,y)$ is given by $f(x,y) = 8xy$ $0 \leq x \leq 1$, $0 \leq y \leq x$
 - (i) Determine the marginal density of X and Y.
 - (ii) Determine the conditional density functions of X and Y. Verify that whether X and Y are independent. [5M CO2 BTL4]

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5. a) Fit a parabola to the following data

X	-2	-1	0	1	2
Y	29	25	22	20	19

[7M CO3 BTL4]

b) Fit a second degree polynomial for the following data by the method of least Squares

x	10	12	15	23	20
y	14	17	23	25	21

[5M CO3 BTL6]

6. a) Categorize random processes into first order ,second order , wide- sense and strict-sense stationary based on their characteristics and properties.

[6M CO4 BTL4]

b) Assess whether the random process $X(t)=A\cos(\omega_0 t+\Theta)$ is wide stationary or not, where A, ω_0 are constants and Θ is a uniformly distributed random variable on the interval $(0,2\pi)$

[6M CO4 BTL5]

7. a) Define and prove any Four properties of PSD of random process. [6M CO5 BTL4]

b) Determine the auto correlation function for the following power density spectrum:

$$S_{xx}(\omega) = \frac{157+12\omega^2}{(16+\omega^2)(9+\omega^2)}$$

[6M CO5 BTL3]

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Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Main) Examinations-January-2023

Electrical Circuit Analysis

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

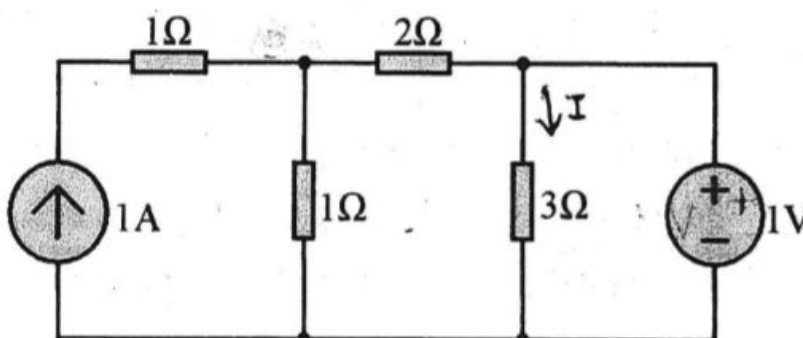
6X2=12M

- a. Define resonance. [2M CO1_ BTL1]
b. State Maximum power transfer theorem [2M CO2_ BTL2]
c. What is time constant? [2M CO3_ BTL2]
d. List the advantages of Laplace Transform methods compared to differential equations. [2M CO4_ BTL1]
e. Define driving Point Function and Transfer Function. [2M CO5_ BTL1]
f. State the Norton's theorem. [2M CO2_ BTL2]

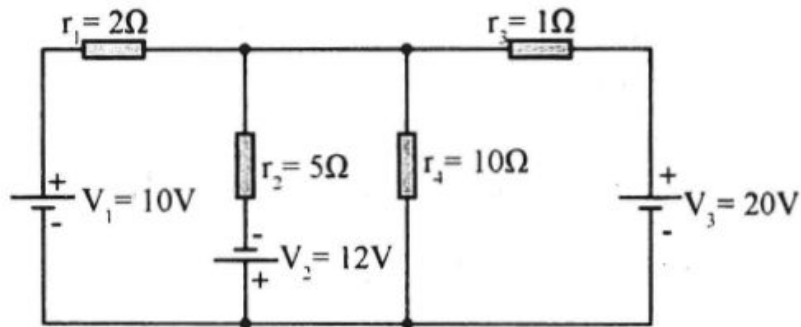
Part-B

4X12=48M

- a) Explain three phase star and delta connected systems and mention the relation between phase and line values of voltage and currents. [6M CO1_ BTL2]
b) Explain the concept of series resonance and derive the formula for resonant frequency. [6M CO1_ BTL3]
- a) Find I in the circuit in 3Ω resistor by using superposition theorem. [6M CO2 BTL3]

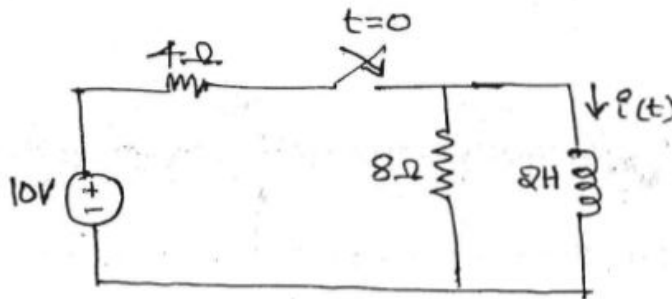


- b) Find the current through the 10Ω resistor using Thevenin's Theorem. [6M CO2 BTL3]



4. a) Find $i(t)$ in an inductor for the circuit shown below.

[6M CO3_BTL4]

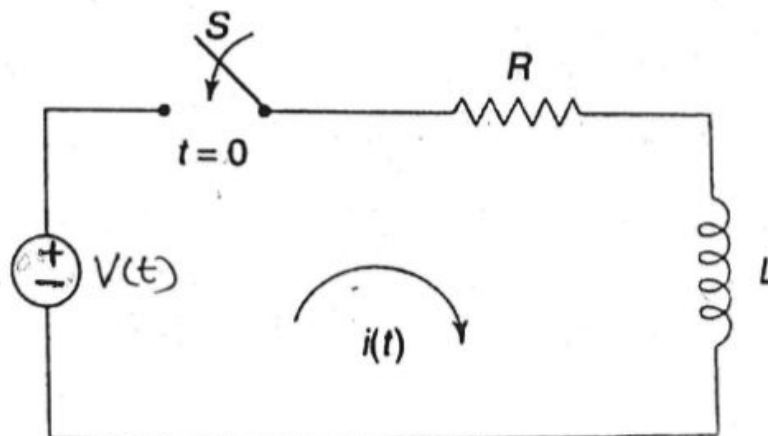


- b) Analyze the RL series circuit with DC excitation and derive the formula for current.

[6M CO3_BTL4]

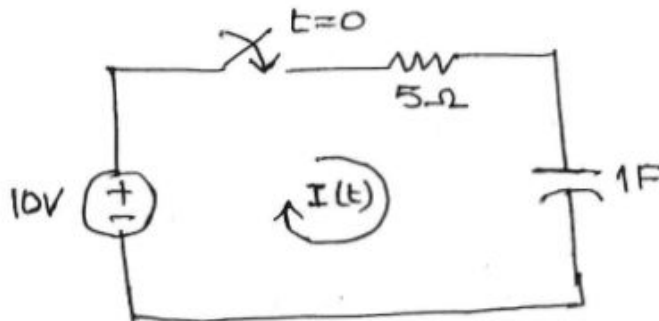
5. a) Find the step response of RL network shown below using Laplace transform method.

[6M CO4_BTL3]



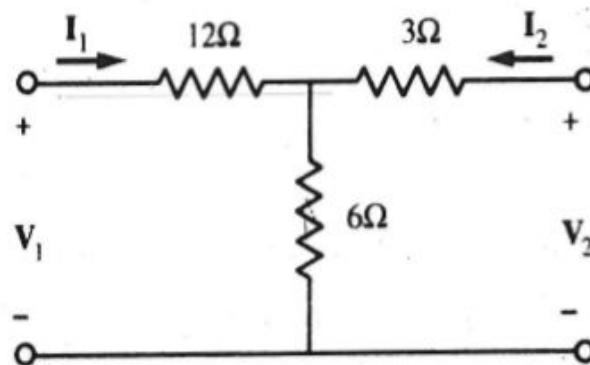
b) Find $I(t)$ for the circuit shown below using Laplace Transform.

[6M CO4_BTL3]



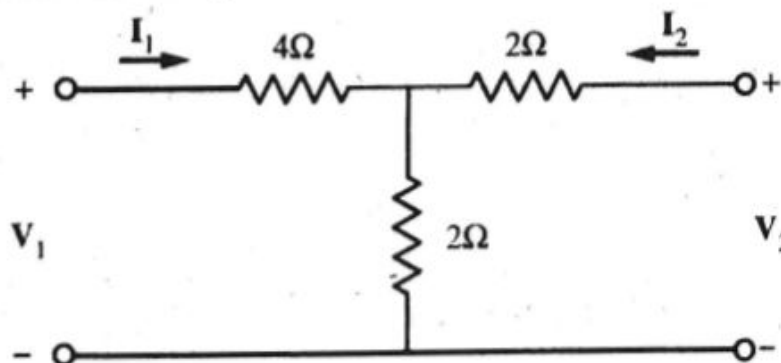
6. a) Determine the z parameters for the circuit in the following figure

[6M CO5_BTL3]



b) Determine the admittance parameters of the network shown below

[6M CO5_BTL3]



7. a) Explain in detail about series and parallel dot convention.

[6M CO1_BTL2]

b) State and explain Reciprocity theorem.

[6M CO2_BTL2]

Stanley College of Engineering & Technology for Women (A)**B.E (EEE) III Semester (Main) Examinations-January-2023****Analog Electronics****Time: 03 Hours****Max. Marks-60**

- Note:** i. First Question is Compulsory. Answer any Four out of remaining Six questions.
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 iii. Missing data, if any, may be suitably assumed.

Part-A**6X2=12M**

1. a. Distinguish between intrinsic & extrinsic semiconductors.

[2M CO1 BTL2]

b. Differentiate Clippers and clampers.

[2M CO2 RTL2]

c. Draw the small signal model of CE amplifier.

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d. Compare between positive and negative feedback.

[2M]

e. List out the types of power amplifier based on conduction angle?

[2M CO4 BTL2]

f. Discuss the concept of virtual ground.

[2M CO5 BTL2]

Part-B**4X12=48M**

2. a) Describe the construction and working of a full wave rectifier.

[6M CO1 BTL2]

b) Demonstrate the formation of depletion region and barrier potential in PN junction diode.

[6M CO1 BTL2]

3. a) Elaborate Input & output characteristics of CE Configuration.

[7M CO2 BTL4]

b) Illustrate the operation of a N-channel JFET.

[5M CO2 BTL3]

4. a) Examine current shunt and voltage series feedback amplifiers?

[6M CO3 BTL3]

b) Derive the expression for positive feedback. Explain Barkhausen criterion.

[6M CO3 BTL3]

5. a) Discuss about the transformer coupled amplifier in detail.

[6M CO2 BTL2]

b) Derive the frequency of Colpitt's oscillator?

[6M CO4 BTL2]

6. a) Describe the working of Wein bridge oscillator. List out its advantages and disadvantages.

[6M CO4 BTL3]

b) Discuss the working of complementary symmetry pushpull power.

[6M CO4 BTL3]

7. a) Illustrate how OP-AMP act as an integrator and differentiator.

[7M CO5 BTL3]

b) Discuss the DC characteristics of an OP-AMP.

[5M CO5 BTL2]



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Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Main) Examinations-January-2023

Electromagnetic Fields

Time: 03 Hours

Max. Marks-60

- Note:**
- First Question is Compulsory. Answer any Four out of remaining Six questions.
 - Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
 - Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

- a. What are the three orthogonal coordinate systems? Give brief [2M CO1 BTL1]
b. Differentiate between divergence and curl. [2M CO1 BTL1]
c. What are the boundary conditions of perfect dielectric materials? [2M CO3 BTL1]
d. Define electric field intensity and electric flux density. [2M CO2 BTL1]
e. Briefly explain about displacement current. [2M CO4 BTL2]
f. What do you understand by Uniform Plane Waves? [2M CO5 BTL2]

Part-B

4X12=48M

- a) What are the types of vector multiplications? And explain each with applications. [6M CO1 BTL1]
b) Calculate the cylindrical coordinates from the given Cartesian coordinates $X=5$, $Y=7$, $Z=8$. [6M CO1 BTL3]
- a) Derive Poisson's and Laplace's equations and write applications. [6M CO2 BTL6]
b) State and prove Gauss's law and write its applications? [6M CO2 BTL2]
- a) State Biot-Savarts law. Differentiate between scalar and vector magnetic potentials. [6M CO3 BTL1]
b) Define Magnetization. Obtain the expression for magnetic flux density in terms of magnetization. [6M CO3 BTL1]
- a) Write the point form of Maxwell's equations. Explain their significance. [6M CO4 BTL2]
b) Explain about Faraday's law for Electromagnetic induction. [6M CO4 BTL2]
- a) State and explain Poynting theorem and what is its significance? [6M CO5 BTL2]
b) Define skin effect and write Maxwell's equation in Phasor form. [6M CO5 BTL2]
- a) Formulate the expression for electric field due to a line charge. [6M CO2 BTL6]
b) Derive the formula for energy density in electric field. [6M CO2 BTL6]

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Stanley College of Engineering & Technology for Women (A)

B.E (EEE) III Semester (Main) Examinations-January-2023

Signals and Systems Analysis

Time: 03Hours

Max. Marks-60

- Note: i. First Question is Compulsory. Answer any Four out of remaining Six questions.
ii. Answer to each question must be written at one place only and in the same order as they occur in the Question paper.
iii. Missing data, if any, may be suitably assumed.

Part-A

6X2=12M

1. a. Define and sketch unit step and unit ramp [2M CO1 BTL1]
b. Define sampling theorem [2M CO1 BTL1]
c. State additivity and homogeneity property [2M CO2 BTL1]
d. List the Fourier transform properties of linearity, time shifting, frequency shifting with expression. [2M CO3 BTL1]
e. Give the differences between DTFT and DFT [2M CO4 BTL2]
f. Find the Laplace Transform of $f(t)=4t^3+t^2-6t+7$ [2M CO5 BTL2]

Part-B

4X12=48M

2. a) Explain about types of signals. [7M CO1 BTL3]
b) Discuss how to reconstruct the signal from its samples in detail. [5M CO1 BTL3]
3. a) Explain the properties of Convolution Integral. [6M CO2 BTL4]
b) Define Linearity, Shift invariance Causality, Stability of LTI System. [6M CO2 BTL2]
4. Find the Trigonometric Fourier Series representation of a periodic square wave
 $x(t)=1$, for the interval $(0, \pi)$; $x(t)=0$, for the interval $(\pi, 2\pi)$ [12M CO3 BTL4]
5. a) State and Prove Parseval's Energy Theorem for discrete signals. [7M CO4 BTL5]
b) Find the discrete time Fourier transform of following
 $x(n)=(0.5)^n u(n)+2^{-n} u(-n-1)$ [5M CO4 BTL4]
6. a) Find the inverse Laplace Transform of the following expression
 $X(S) = 3S^2 + 8S + 6 / (S+2)(S^2+2S+1)$ [5M CO5 BTL4]
b) Find the Laplace Transform of the function
 $f(t)=3t^4-2t^3+4e^{-3t}-2\sin 5t+3\cos 2t$ [7M CO5 BTL4]
7. a) Define the Z transform and explain ROC of finite duration sequences in detail. [7M CO5 BTL2]
b) If $x(n)=x_1(n)*x_2(n)$, where $x_1(n)=(1/3)^n u(n)$ and $x_2(n)=(1/5)^n u(n)$. Find $X(z)$ by using convolution property for z-transform. [5M CO5 BTL4]

Stanley College of Engineering and Technology for Women(A)**B.E(EEE,ECE & IT) II Semester (Backlog) Examinations January-2023****Indian Constitution**

Time: 3 hours

Max.Marks:60

PART-A**5 X 2=10M****Note: Answer all questions(Compulsory)**

- | | |
|---|------------|
| 1 Who are the members of Drafting Committee? | 2 M CO1 L1 |
| 2 What are Panchayati Raj Institutions?Name them. | 2 M CO2 L2 |
| 3 Explain about Article-19. | 2 M CO3 L2 |
| 4 Why is NITI Aayog Necessary? | 2 M CO4 L1 |
| 5 What is the Composition of National Commission for Women? | 2 M CO5 L3 |

PART-B**5 X 10= 50M****Note: Answer all questions**

- | | |
|---|------------|
| 6 a. Discuss the Significance of Indian Constitution. | 5 M CO1 L4 |
| b. What are the Features of 1935 Act? | 5 M CO1 L1 |

OR

- | | |
|--|-------------|
| c. Explain the Philosophical Foundations to the Indian Constitution. | 10 M CO1 L3 |
| 7 a. Examine the Powers and Functions of Prime Minister. | 5 M CO2 L4 |
| b. Explain about Emergency powers of President of India. | 5 M CO2 L2 |

OR

- | | |
|--|-------------|
| c. Write about Evolution of Local Government in India. | 10 M CO2 L4 |
| 8 a. Explain the Importance of Fundamental duties. | 5 M CO3 L3 |
| b. Briefly explain about DPSP. | 5 M CO3 L4 |

OR

- | | |
|---|------------|
| c. Explain the features of Fundamental rights. | 5 M CO3 L2 |
| d. Explain the Five Writs in Indian Constitution. | 5 M CO1 L1 |
| 9 a. Explain about Composition of Inter State Council. | 5 M CO4 L3 |
| b. Write an essay on Role and Functions of Finance commission of India. | 5 M CO4 L4 |

OR

- | | |
|---|-------------|
| c. Explain about the functions of NITI Aayog. | 5 M CO4 L2 |
| d. Explain about Administrative relations of Union-State. | 5 M CO4 L4 |
| 10 a. What are the Powers and Functions of NHRC? | 10 M CO5 L4 |

OR

- | | |
|--|-------------|
| c. What are the Powers and Functions of National Commission for Women? | 10 M CO5 L1 |
|--|-------------|

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Stanley College of Engineering and Technology for Women(A)

B.E (ECE, EEE & IT) II Semester (Backlog) Examinations January-2023

Essence of Indian Traditional Knowledge

Time: 3 hours

Max.Marks:60

PART-A

Note:Answer all questions (Compulsory)

5 X 2=10M

- 1 Define Civilization. 2M L1
- 2 What are Brahmanas and Aryankas? 2M L1
- 3 List the Heterodox schools of Indian Philosophy. 2M L1
- 4 Write a short note on Sama Veda. 2M L1
- 5 What were the subjects of study in Ancient India? 2M L1

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Discuss the general characteristics of Culture. 5M L2
b. Explain the uniqueness of Indian Culture. 5M L2
OR
c. Write about ancient India. 5M L1
d. Write an essay on the influence of the English in Modern India. 5M L1
- 7 a. Summarize the developments of South Indian Languages and Literature. 5M L2
b. Explain about the Vedas and Upanishads. 5M L2
OR
c. Discuss the classic Sanskrit literature. 5M L2
d. Discuss the literature of North Indian Languages. 5M L2
- 8 a. Examine the salient features of orthodox schools of Indian Philosophy. 5M L2
b. What is the important role played by the Bhakti and Sufi religious reform movements in India? 5M L2
OR
c. What is Sunyavada of Buddhism?Explain. 5M L2
d. Discuss the Religious reform movements in Modern India. 5M L2
- 9 a. Write an essay on "Ancient Indian Architecture". 5M L2
b. Explain elaborately about the Fine Arts of India. 5M L2
OR
c. Discuss about Indian classical dance forms. 5M L2
d. Justify the statement "Science and technology can solve all the world's problems, and historically it has been shown to make the world better and better." 5M L3
- 10 a. Write about the education system in Medieval India. 5M L1
b. List any five contributions made by Ancient Indian scientists to the world of Science. 5M L1
OR
c. Explain the Gurukula Education system. 5M L2
d. Discuss the role of Education in social change. 5M L1

Hall Ticket No-

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Stanley College of Engineering and Technology for Women(A)

B.E (ECE & EEE) II Semester (Backlog) Examinations January-2023

Engineering Physics

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 A plane transmission grating having 6000 lines per cm is used in normal incidence mode. Calculate the angular separation between the two spectral lines of wavelengths 5890 Å and 5896 Å in second order. 2 M
- 2 Explain the phenomenon of total internal reflection. 2 M
- 3 Define a) Space lattice and b) Unit cell. 2 M
- 4 Calculate the wavelength of an electron accelerated to a potential of 900 V. 2 M
- 5 What is Lenz's law? 2 M

PART-B

Note: Answer all questions

5 X 10= 50M

- 6 a. Explain the Young's double slit experiment to produce the interference pattern and obtain the conditions for maxima and minima. 5 M
b. Discuss intensity expression for N slit diffraction grating. 5 M
OR
c. Analyze interference of light by division of amplitude and division of wave front with examples. 5 M
d. Classify Fresnel's and Fraunhofer's diffraction patterns. 5 M
- 7 a. Explain the construction and working of semiconductor laser. 5 M
b. Explain the types of optical fibers basing on refractive index profiles. 5 M
OR
c. Analyze the basic components required for producing laser 5 M
d. Derive an expression for Numerical aperture and acceptance angle of an optical fiber. 5 M
- 8 a. Deduce the equation for the interplanar distance for cubic crystal. 5 M
b. Classify the point defects in a crystal. 5 M
OR
c. Formulate the expression for the concentration of Schottky defects in ionic crystals. 5 M
d. What are Miller indices? How to obtain them for a set of lattice planes? 5 M
- 9 a. Formulate the Schrodinger's time independent wave equation. 5 M
b. Explain polarization in dielectric materials. 5 M
OR
c. Summarize the properties of wave function Ψ and explain its significance. 5 M
d. Determine an expression for electric field due to continuous charge distribution. 5 M
- 10 a. Explain the concept of displacement current in dielectrics. 5 M
b. Explain Faraday's law and Lenz's law. 5 M
OR
c. Obtain Poynting theorem. 5 M
d. Build the Maxwell's equations for static fields. 5 M

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Stanley College of Engineering and Technology for Women (A)

B.E(ECE,EEE & IT) II Semester (Backlog) Examinations January-2023

English

Time : 3 hours

Max.Marks : 60

PART-A

Note : Answer all questions (Compulsory)

5 X 2=10M

2 M BL1 CO1

- 1 Select the appropriate answer and fill in the blanks . (1/2m x 4 = 2 M)
 - a. She does all her work _____ great care. (with/ by)
 - b. I saw myself _____ the mirror. (in/on)
 - c. The necklace was _____ present from her parents. (a/an/the)
 - d.They were going to _____ concert. (Article) (a/ an/ the)
- 2 Use the appropriate tense forms of the given verbs and fill in the blanks.(1/2 m x4 = 2M) 2 M BL1 CO2
 1. How have you been _____ ? (do)
 2. He was badly wounded and _____ (bleed) profusely.
 - 3 .I usually _____ (teach) Chemistry, but I _____ (teach) Biology this year.
 - 4.She _____(finish) her breakfast and went to school.
- 3 Select the correct answer. (1/2m x 2 = 1 M) 2 M BL4 CO3
 - a. A study of coins. _____ (numismatics/balistics)
 - b. Handwriting which is difficult to read. _____ (legible/illegible)Rewrite as directed.(1/2m x 2 = 1 M)
 - c. Lavanya said, " We painted the house yesterday (Indirect speech)."
 - d. She said, "I gave the key to my roommate (Indirect Speech)."
- 4 Select the appropriate word and fill in the blanks. (1/2mx 2 = 1M) 2 M BL1 CO4
 - a. Let us _____ to the auditorium. (precede, proceed)
 - b. The bus _____ (fare/fair) increased drastically.Change the voice. (1/2m x 2 = 1M)
 - c.The students have paid the fees.
 - d. She was congratulated by her friends.
- 5 Rewrite the sentences correctly (1/2m x 4 = 2 M) 2 M BL3 CO5
 - a. I will look forward to see you soon.
 - b. Please don't repeat the mistake again.
 - c. The water is very hot to drink.
 - d. Harsha don't know Hindi.

PART-B

Note : Answer all questions

5 X 10= 50M

- 6 a. Summarize " On the Conduct of Life" by William Hazlitt briefly. 5 M BL2 CO1
- b. Use the following hints to write a paragraph, give it a title and a moral. 5 M BL3 CO1
Hints : The King of Scotland was defeated six times-----hid himself in a cave saw a spider trying to climb up to the roof of the cave the spider fell six times and then succeeded- the king was encouraged - tried again to defeat his enemies and succeeded.

OR

Code :S222821

- c. Summarize the young boy's attitude towards his new school as explained in William Hazlitt's "On the Conduct of Life." 5 M BL2 CO1
- d. Examine the structure of writing an effective paragraph. 5 M BL3 CO1
- 7 a. Outline the central idea of the poem "If." 5 M BL2 CO2
- b. Describe compounding, clipping and blending words with examples. 5 M BL1 CO2
- OR
- c. Outline the poet's views on honesty and fortitude of character from the poem "If." 5 M BL1 CO2
- d. Describe briefly "Recent floods due to continuous rains." 5 M BL2 CO2
- 8 a. Discuss the dream that Martin Luther King envisions for America. 5 M BL2 CO3
- b. Describe your childhood memories. 5 M BL2 CO3
- OR
- c. Discuss "I have a Dream" as a plea for freedom and equality. 5 M BL2 CO3
- d. Describe your views on "Is country life better than city life." 5 M BL2 CO3
- 9 a. Examine the conflict of the Poem "The Road not Taken." 5 M BL3 CO4
- b. Justify the topic "Should the minimum voting age be raised/lowered/kept the same?" 5 M BL5 CO4
- OR
- c. Examine the spirit of regret and longing in the poem "The Road not Taken." 5 M BL3 CO4
- d. Justify the topic "Is online learning effective?" 5 M BL5 CO4
- 10 a. Explain the important features of a blog. 5 M BL2 CO5
- b. Create a blog on the topic: "Writing captions on instagram." 5 M BL6 CO5
- OR
- c. Explain the process of writing "Tweets" and "Instagrams." 5 M BL2 CO5
- d. Create a tweet in response to the proposal of the "New ILabour Code.i.e. 4- day work per week, 12 working hours every working day in the week." 5 M BL6 CO5

Hall Ticket No-

Code S222825

Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) II Semester (Backlog) Examinations January-2023

Mathematics--II

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5X2=10M

- 1 Solve $(x^2 + y^2) dx + 2xy dy = 0$ 2 M L3
- 2 Find the C.F. of $(D^2 + D + 1)y = 0$ 2 M L2
- 3 Define Beta function and Gamma function 2 M L1
- 4 Find the Eigen Values of the Matrix $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ 2 M L3
- 5 Find $L\{t^2 \sin t\}$ 2 M L3

PART-B

Note: Answer all questions

5X10=50M

- 6 a. Solve $\frac{dy}{dx} + y \cos x = y^3 \sin 2x$ 5 M L3
- b. Find the Orthogonal trajectories of $\frac{x^2}{a^2 + \alpha} + \frac{y^2}{b^2 + \alpha} = 1$, α being the parameter. 5 M L3
- c. Solve $(1 + e^{xy}) dx + e^{xy} (1 + \frac{x}{y}) dy = 0$ 5 M L3
- d. Solve $p = \sin(y + xp)$. Also find its singular solutions. 5 M L3

OR

- 7 a. Solve $(D^2 + 4)y = e^x + \sin 2x + \cos 2x$ 5 M L3
- b. Solve $y'' + y = \sec x$ by the Method of Variation of parameters. 5 M L3
- c. Solve $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} + 2y = x \log x$ 5 M L3
- d. Solve $(D^2 - 4D + 3)y = 3e^x \cos 2x + 2xe^{3x}$ 5 M L3

OR

- 8 a. State and prove the relation between Beta and Gamma Function. 6 M L4
- b. Evaluate $\int_0^2 x^3 \sqrt{2-x} dx$ using Beta And Gamma function. 4 M L5

OR

- c. State and Prove Rodrigue's Formula. 6 M L4
- d. Evaluate $\int_0^a x^4 \sqrt{a^2 - x^2} dx$ 4 M L5

Code S222825

- 9 a. Verify Cayley – Hamilton Theorem for the Matrix 6 M L4
 $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and also find A^{-1}

- b. Discuss the nature of the Q.F. $x^2 + 4xy + 6xz + 2yz - y^2 + 4z^2$ 4 M L4

OR

- c. Find Eigen Values and corresponding Eigen Vectors of 8 M L3
 $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

- d. Using Cayley – Hamilton Theorem, find A^8 if $A = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}$ 2 M L3

- 10 a. Find the Laplace Transform of (i) $t e^{2t} \sin 3t$ (ii) $e^{-t} \cos^2 t$ 5 M L3

- b. Find the Inverse Laplace Transform of $\frac{2s+12}{s^2+6s+12}$ and $\frac{s}{(s+2)^2}$ 5 M L3

OR

- c. Find $L \{ t \cos at \}$ 2 M L5

- d. Using Laplace Transform method, solve 8 M L5
 $(D^2 + 1)y = 6 \cos 2t, t > 0$, If $y = 3, Dy = 1$ when $t = 0$.

Hall Ticket No.

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Stanley College of Engineering and Technology for Women (A)

B.E (CSE/EEE/ECE/IT/CME/AI&DS) I Semester (Backlog) Examinations January-2023

Mathematics-I

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

1. Test the convergence of the series $\sum (1 + \frac{1}{n})^{-n^2}$ 2M
2. Verify Cauchy's mean value theorem for $f(x) = x^2$ and $g(x) = x^3$ in $[1, 2]$ 2M
3. If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1} x + \tan^{-1} y$ then prove that u and v are functionally dependent. 2M
4. Find grad f if $f = x^2y + y^2x + z^2$ at $(1, 1, -2)$ 2M
5. Find the first approximation of the root lying between 0 and 1 of the equation $x^3 + 3x - 1 = 0$ by Newton-Raphson formula. 2M

PART B

Note: Answer all questions

5 X 10 = 50M

6. (a) Discuss the convergence or divergence of $\frac{x}{1.2} + \frac{x^2}{2.3} + \frac{x^3}{3.4} + \dots$ 5M
 - (b) Examine the convergence of the series $1 - \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{4}} + \dots$ 5M
- (OR)

- (c) Test the convergence of the series $1 + \frac{2}{5}x + \frac{6}{9}x^2 + \dots + \frac{2^n-2}{2^{n+1}}x^{n-1} + \dots$ ($x > 0$) 5M
 - (d) Show that the exponential series $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ converges absolutely for all values of x . 5M
7. (a) Find the evolute of the parabola $y^2 = 4ax$ 5M
 - (b) Verify Cauchy's Mean value theorem for the functions $f(x)$ and $f^1(x)$ in $(1, e)$ given $f(x) = \log x$ 5M

(OR)

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(c) Prove that $\frac{\pi}{6} + \frac{1}{5\sqrt{3}} < \sin^{-1}\left(\frac{3}{5}\right) < \frac{\pi}{6} + \frac{1}{8}$ 10M

8. (a) If $u = x + y + z$, $y + z = uv$, $z = uvw$ then show that $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$ 5M

(b) If $u = x^2 + y^2$, $x = at^2$, $y = 2at$ then find $\frac{du}{dt}$ 5M

(OR)

(c) A rectangular box open at the top is to have volume 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 5M

(d) Find the maximum and minimum values of $f(x,y) = x^3 + y^3 - 3axy$ 5M

9. (a) If $F = (5xy - 6x^2)\mathbf{i} + (2y - 4x)\mathbf{j}$ Evaluate $\int_c \vec{F} \cdot d\vec{r}$ where c is the curve in the xy plane $y = x^3$ from $(1,1)$ to $(2,8)$ 5M

(b) Find $\text{div } \vec{f}$ when $\vec{f} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$ 5M

(OR)

(c) Verify Green's theorem in the plane for $\oint (3x^2 - 8y^2)dx + (4y - 6xy)dy$ where C is the region bounded by $y = \sqrt{x}$ and $y = x^2$ 10M

10. (a) Solve the following system of equations using Gauss Seidel iteration method

$$10x + 2y + z = 9, \quad x + 10y - z = -22, \quad -2x + 3y + 10z = 22 \quad 10M$$

(OR)

(c) Using Newton's divided difference formula evaluate $f(8)$ given that 10M

x	4	5	7	10	11	13
f(x)	48	100	294	900	1210	2028

Stanley College of Engineering and Technology for Women(A)**B.E(ECE & EEE) I Semester (Backlog) Examinations January-2023****Engineering Chemistry****Time: 3 hours****Max.Marks:60****PART-A****Note: Answer all questions (Compulsory)****5 X 2=10M**

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|---|------------|
| 1 What is cathodic protection? | 2 M L1 CO1 |
| 2 Define entropy and enthalpy. | 2 M L2 CO2 |
| 3 Calculate the bond order of NO molecule. | 2 M L3 CO3 |
| 4 Describe the significance of Octane number and Cetane number of a fuel. | 2 M L4 CO4 |
| 5 What are the composite materials? Write its classification. | 2 M L2 CO5 |

PART-B**Note: Answer all questions****5 X 10= 50M**

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|---|------------|
| 6 a. Explain the purification of water by ion exchange process and give its advantages,disadvantages. | 6 M L2 CO1 |
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| b. Discuss the various (any three) factors affecting rate of corrosion. | 4 M L6 CO1 |
|---|------------|

OR

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| c. What are boiler troubles? Explain formation & prevention of scale and sludge. | 6 M L2 CO1 |
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| d. Write a short note on galvanizing. | 4 M L1 CO1 |
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| 7 a. Explain the physical significance of entropy. | 6 M L2 CO2 |
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| b. Derive nernst equation and write its applications. | 4 M L6 CO2 |
|---|------------|

OR

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| c. Describe the construction and working of calomel electrode with a neat diagram. | 6 M L5 CO2 |
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| d. The efficiency of an engine is 0.42. Calculate the heat that must be withdrawn from the reservoir at higher temperature to produce 203 cal of work. | 4 M L3 CO2 |
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| 8 a. Draw the molecular orbital diagram of N ₂ and find out the bond order. | 6 M L4 CO3 |
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| b. Describe the preparation and applications of Poly lactic acid. | 4 M L2 CO3 |
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OR

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| c. Discuss the preparation,properties and applications of PET and Kevlar | 6 M L6 CO3 |
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| d. Analyze the doping mechanism of conduction in poly acetylene(p-doping & n-doping) | 4 M L4 CO3 |
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| 9 a. Explain proximate analysis of coal? How is it carried out ? What is its significance? | 6 M L1 CO4 |
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| b. A sample of coal has the following composition: C=83% ,H=7.5% , S=1.5% , N=0.6 % ,O=8.4 % find the gross calorific value using Dulong Formula. | 4 M L3 CO4 |
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OR

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| c. Explain fractional distillation of petroleum with a neat diagram. | 6 M L2 CO4 |
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| d. Explain the concept of trans esterification of biodiesel. | 4 M L2 CO4 |
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| 10 a. What are Lithium ion batteries? Explain the construction,working and applications of Lithium ion batteries. | 6 M L5 CO5 |
|---|------------|

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| b. Explain the construction and working of CH ₃ OH-O ₂ fuel cell and write its advantages. | 4 M L2 CO5 |
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OR

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| c. Illustrate the principles of green chemistry. | 6 M L3 CO5 |
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| d. Make the use of Nanotechnology write a short note on the following a) Carbon nano tubes b) Nano crystals. | 4 M L3 CO5 |
|--|------------|

Stanley College of Engineering and Technology for Women(A)

B.E(EEE & ECE) I Semester Backlog Examinations-January-2023

Fundamentals of Electrical Engineering

Time: 3 hours

Max.Marks:60

Part-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- 1 State Kirchhoff's current law. 2 M
- 2 Define Reactive Power and Apparent Power. 2 M
- 3 What are the differences between statically induced emf & dynamically induced emf? 2 M
- 4 Why single-phase induction motor is not self-starting? 2 M
- 5 Mention any two components of LT Switchgear. 2 M

Part-B

Note: Answer all questions

5 X 10= 50M

- 6
 - a. Formulate the expressions for equivalent resistance if 'n' number of resistances is connected in series and parallel combinations. 5 M
 - b. Explain Thevenin Theorem with one example. 5 M
- OR**
- c. Explain Norton Theorem with one example. 5 M
 - d. What is Superposition Theorem, Justify with an example? 5 M
- 7
 - a. Derive the expressions for resistances when a network is converted from star to delta. 5 M
 - b. The equation of an alternating current is $I=42.42 \sin 628t$. Determine (i) its maximum value (ii) Frequency (iii) RMS value (iv) Average value and (v) Form Factor. 5 M
- OR**
- c. Explain the behavior of a pure inductor when connected across a single-phase ac supply. 5 M
 - d. Show that the line voltage is $\sqrt{3}$ times the phase voltage in a star connected 3- ϕ system. 5 M
- 8
 - a. Explain the principle of operation of a three-phase Induction Motor. 5 M
 - b. A 10 KVA, 500/250V single phase transformer has the following results
 - i) O.C. Test : 250V , 0.8A, 200W
 - ii) S.C. Test : 120V, 20A, 500W
 Calculate the Efficiency at Full load with 0.8P.F. lagging 5 M

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- c. Explain the principle of operation of a Transformer. 5 M
- d. Discuss in detail about the Rotating Magnetic Field in a 3-phase Induction Motor. 5 M

- 9
- a. With a neat circuit diagram, Discuss about the Split-phase type Induction Motor 5 M
 - b. Derive an expression for the EMF Equation of a DC Generator. 5 M

OR

- c. A 6-pole machine has an armature with 90 slots and 8 conductors per slot, the flux per pole is 0.05 wb and speed at 1000 rpm. Determine induced emf if winding is (i) lap connected and (ii) wave connected. 5 M
- d. Draw a neat sketch of a DC Machine and explain the function of various parts. 5 M

- 10
- a. Explain the concept of earthing. 5 M
 - b. Clearly illustrate types of wires and cables. 5 M

OR

- c. Explain in detail MCB and ELCB. 5 M
- d. Mention methods of power Factor improvement. 5 M
